A RELATIVE DUALIZING COMPLEX RESTRICTS ON AN AFFINE CHART TO THE ALGEBRAIC RELATIVE DUALIZING COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E2U (sources/stacks-duality/pinned-original-source.tex line 7271).

Allowed standard prerequisites (editorial).

- The affine comparison between complexes of modules and complexes of sheaves. Derived Categories of Schemes, Lemma labelled perfect-lemma-affine-compare-bounded, cited in the statement. Identifies the restriction of K with an object of D(A).
- Compatibility of pushforward, pullback and internal Hom with the affine comparison. Derived Categories of Schemes, Lemmas labelled perfect-lemma-quasi-coherence-pushforward, perfect-lemma-quasi-coherence-pullback and perfect-lemma-quasi-coherence-internal-hom, cited at proof lines 7326-7336. Translate the sheaf conditions into algebraic ones.
- Pseudo-coherence of A over A otimes_R A for the diagonal. More on Algebra, Lemma labelled more-algebra-lemma-more-relative-pseudo-coherent-is-moot, cited at proof lines 7340-7344. Supplies the pseudo-coherence needed for the internal Hom comparison.
- The algebraic definition of a relative dualizing complex. Dualizing Complexes, Definition labelled dualizing-definition-relative-dualizing-complex, cited in the statement and at proof lines 7344-7348. Fixes the conditions recovered at the end.

Source reference. Original lemma 06Z0: sources/stacks-chow/tags.txt, source lines 8904–8904. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original definition 0E2C: sources/stacks-chow/tags.txt, source lines 17602–17602. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7314 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08E5: sources/stacks-chow/tags.txt, source lines 10661–10661. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7295 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0DJK: sources/stacks-chow/tags.txt, source lines 16980–16980. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7301 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08DW: sources/stacks-chow/tags.txt, source lines 10652–10652. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7302 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A6H: sources/stacks-chow/tags.txt, source lines 12845–12845. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7303 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E1T: sources/stacks-chow/tags.txt, source lines 17583–17583. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7310 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $X \to S$ be a morphism of schemes which is flat and locally of finite presentation. Let (K, ξ) be a relative dualizing complex. Then for any commutative diagram*

$$\begin{array}{ccc} \mathrm{Spec}(A) & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(R) & \longrightarrow & S \end{array}$$

whose horizontal arrows are open immersions, the restriction of K to $\mathrm{Spec}(A)$ corresponds via Derived Categories of Schemes, Lemma 06Z0 to a relative dualizing complex for $R \to A$ in the sense of Dualizing Complexes, Definition 0E2C.

Proof. Since formation of $R\mathcal{H}om$ commutes with restrictions to opens we may as well assume $X = \mathrm{Spec}(A)$ and $S = \mathrm{Spec}(R)$. Observe that relatively perfect objects of $D(\mathcal{O}_X)$ are pseudo-coherent and hence are in $D_{QCoh}(\mathcal{O}_X)$ (Derived Categories of Schemes, Lemma 08E5). Thus the statement makes sense. Observe that taking Δ_*, $L\mathrm{pr}_1^*$, and $R\mathcal{H}om$ is compatible with what happens on the algebraic side by Derived Categories of Schemes, Lemmas 0DJK, 08DW, 0A6H. For the last one we observe that $L\mathrm{pr}_1^*K$ is S-perfect (hence bounded below) and that $\Delta_*\mathcal{O}_X$ is a pseudo-coherent object of $D(\mathcal{O}_W)$; translated into algebra this means that A is pseudo-coherent as an $A \otimes_R A$-module which follows from More on Algebra, Lemma

0E1T applied to $R \to A \otimes_R A \to A$. Thus we recover exactly the conditions in Dualizing Complexes, Definition 0E2C. $\square$